

Awards

- 2024 **Euclid STAR Prize — Team Technical/Engineering Award.**
Team led by Saeeda Awan and Stefano Dusini. Contribution:
NISP filter wheel alignment and optical centering checks during
the commissioning phase.

Software

- psfcraft PSF simulation for the NISP instrument [repo]
zernet Zernike coefficient extraction from pure (noiseless) PSFs [repo]
spread Zernike coefficient extraction from noisy PSFs [repo]

Publications

- 2026 **Spread: a framework for PSF-based aberration analysis.**
Journal of Optics & Laser Technology. doi:10.2139/ssrn.6544752
- 2025 **Euclid: I. Overview of the Euclid mission.** *Astronomy & Astrophysics*. doi:10.1051/0004-6361/202450810
- 2024 **Performance of the image persistence model for Euclid infrared detectors.** . doi:10.1117/12.3014312
- 2024 **Euclid near infrared spectrometer and photometer instrument NISP in space.** . doi:10.1117/12.3017391
- 2024 **Euclid commissioning results: the near infrared spectrometer and photometer (NISP) signal detection chain.** . doi:10.1117/12.3017952
- 2024 **Decoding optical aberrations of low-resolution Instruments from PSFs: machine learning and Zernike polynomials perspectives.** *Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave*. doi:10.1117/12.3018334